

Solution Requirements

Date	10 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Define the functional and non-functional requirements for the Flood Prediction System. These requirements describe the features, system behavior, performance expectations, security, and quality standards needed to develop an accurate, reliable, and user-friendly machine learning application for flood prediction and early warning.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>No user authentication is required. The application is publicly accessible for prediction purposes.</i>	High
2	Authorization levels	<i>All users have equal access to enter weather data and obtain flood predictions.</i>	High
3	External interfaces	<i>Integrate weather datasets and optionally connect to live Weather APIs for real-time rainfall information.</i>	High

4	Transactions processing	<i>Accept weather parameters from users, preprocess the data, generate flood predictions using the trained ML model, and display the prediction result.</i>	High
5	Reporting	<i>Generate flood prediction results, prediction history, accuracy reports, and visual charts for analysis.</i>	Medium
6	Business rules, etc.	<i>Predict flood risk only after validating all required weather parameters. Display Flood Likely or No Flood based on the trained machine learning model.</i>	High
7	Compliance to laws or regulations	<i>Protect user information and comply with applicable data privacy and cybersecurity guidelines while handling weather and prediction data.</i>	Medium
8	Other	<i>Allow future integration with SMS/Email alert systems and live weather forecasting services.</i>	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>Generate flood predictions quickly after receiving user input.</i>	<i>Prediction generated in less than 3 seconds.</i>

2	Scalability	<i>Support increasing numbers of users and larger weather datasets without affecting performance.</i>	<i>Supports 1,000+ prediction requests per day.</i>
3	Security & Data Privacy	<i>Secure user information and prediction data using authentication and encrypted communication where applicable.</i>	<i>Secure login and HTTPS communication.</i>
4	Reliability & Availability	<i>Ensure the application remains available during normal operation and handles prediction requests reliably.</i>	<i>99% system availability.</i>
5	Usability & Accessibility	<i>Provide an intuitive, responsive, and user-friendly interface that works across desktops and mobile devices.</i>	<i>Easy navigation with responsive web design.</i>
6	<i>Other</i>	<i>Maintain modular, well-documented, and easily maintainable code for future improvements and deployment.</i>	Clean project structure with reusable code modules.